

# Mevin B. Hooten

mevin.hooten “at” austin “dot” utexas “dot” edu  
<https://mhooten.github.io/>

## Work Address:

2317 Speedway, D9800  
Austin TX, 78712

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| <b>Experience:</b> | UNIVERSITY OF TEXAS at AUSTIN, Austin, TX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |
|                    | <ul style="list-style-type: none"><li>• Professor2021-<ul style="list-style-type: none"><li>◦ Department of Statistics and Data Sciences</li></ul></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |
|                    | COLORADO STATE UNIVERSITY, Fort Collins, CO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |
|                    | <ul style="list-style-type: none"><li>• Professor2018-2021<ul style="list-style-type: none"><li>◦ Department of Fish, Wildlife, and Conservation Biology</li><li>◦ Department of Statistics</li></ul></li><li>• Associate Professor2013-2018<ul style="list-style-type: none"><li>◦ Department of Fish, Wildlife, and Conservation Biology</li><li>◦ Department of Statistics</li></ul></li><li>• Assistant Professor2010-2013<ul style="list-style-type: none"><li>◦ Department of Fish, Wildlife, and Conservation Biology</li><li>◦ Department of Statistics</li></ul></li><li>• Faculty Affiliate2011-2021<ul style="list-style-type: none"><li>◦ Graduate Degree Program in Ecology</li></ul></li><li>• Assistant Unit Leader2010-2021<ul style="list-style-type: none"><li>◦ Colorado Cooperative Fish and Wildlife Research Unit</li></ul></li></ul> |           |
|                    | UTAH STATE UNIVERSITY, Logan, UT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2006-2010 |
|                    | <ul style="list-style-type: none"><li>• Assistant Professor of Statistics, Department of Mathematics and Statistics</li><li>• Adjunct Faculty, Department of Wildland Resources</li><li>• Faculty Associate, Ecology Center</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |
| <b>Education:</b>  | UNIVERSITY OF MISSOURI, Columbia, MO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2006      |
|                    | <ul style="list-style-type: none"><li>• Ph.D. Statistics Advisor: Christopher K. Wikle</li><li>• Dissertation Topic: Hierarchical spatio-temporal models for ecological processes</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |
|                    | UNIVERSITY OF MISSOURI, Columbia, MO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2001      |
|                    | <ul style="list-style-type: none"><li>• M.S. Forest Ecology Advisor: David R. Larsen</li><li>• Thesis Topic: Modeling the spatial distribution of ground flora</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |
|                    | KANSAS STATE UNIVERSITY, Manhattan, KS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1999      |
|                    | <ul style="list-style-type: none"><li>• B.S. Natural Resource Management Advisor: Mark Morgan</li><li>• Minor in Wildlife Biology</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |

## Major Honors:

- Elected Fellow of the American Statistical Association (2017)
- Distinguished Achievement Award, American Statistical Association, ENVR Section (2022)
- Early Investigator Award, American Statistical Association, ENVR Section (2014)

## Books:

Hobbs, N.T. and M.B. Hooten. (2025). Bayesian Models: A Statistical Primer for Ecologists, Second Edition. Princeton University Press.

Hooten, M.B. and T.J. Hefley. (2019). Bringing Bayesian Models to Life. Chapman & Hall/CRC.

Hooten, M.B., D.S. Johnson, B.T. McClintock, and J. Morales. (2017). Animal Movement: Statistical Models for Telemetry Data. Chapman & Hall/CRC.

Hobbs, N.T. and M.B. Hooten. (2015). Bayesian Models: A Statistical Primer for Ecologists. Princeton University Press.

## Selected Publications: (students and post-docs underlined)

Eisaguirre, J.M., P.J. Williams, T.J. Hefley, and M.B. Hooten. (In Press). Ecological diffusion models are still useful in ecology. *Landscape Ecology*.

Arab, A., M.B. Hooten, and C.K. Wikle (In Press). Hierarchical Spatial Models. *In: Encyclopedia of Geographical Information Science, Third Edition*. Springer.

Glazer, A.K., L. Parast, and Hooten, M.B. (In Press). Beyond the yard line: Accommodating rounded sports data in statistical models. *The American Statistician*.

Van Ee, J.J., D. Gamba, J.R. Lasky, M.L. Vahsen, and M.B. Hooten. (In Press). Spatial knockoff Bayesian variable selection in genome-wide association studies. *Bayesian Analysis*.

Vahsen, M.L., J.J. Van Ee, D. Gamba, T.M. Maxwell, N. Pirtel, S.J. Romero, D.R. Barnett, O. Baughman, D.J. Ensing, R. Gill, M.C. Holdrege, R.A. Hufbauer, R. Hufft, C.E. Moffat, J. Ott, L. Pyle, C. Schroeder, E.W. Schupp, R.K. Shriver, M. Stemkovski, A.J. Symstad, A.K. Urza, D.M. Blumenthal, M.J. Germino, M.B. Hooten, J.R. Lasky, E.A. Leger, L.M. Porensky, and P.B. Adler. (2026). Eco-evolutionary context modifies a destructive plant invader's response to climate. *New Phytologist*, **251**: 2980-2994.

Glazer, A.K., L. Parast, and M.B. Hooten. (2026). Beyond the yard line: Accommodating rounded sports data in statistical models. *The American Statistician*, **80**: 443-452.

Van Ee, J.J., Y. Kanno, X. Lu, J.M. Rash, and M.B. Hooten. (2026). Geometric overdispersion facilitates the integration of ecological data. *Methods in Ecology and Evolution*, **17**: 2104-2122.

Van Ee, J.J., M.L. Vahsen, D. Gamba, T. Maxwell, B.E. Davidson, B.E. Lazarus, L. Porensky, R. Hufbauer, M.J. Germino, P.B. Adler, J.R. Lasky, and M.B. Hooten. (2026). Data reconciliation in multi-trait experiments with kinship ordination. *Methods in Ecology and Evolution*, **17**: 1799-1817.

Koslovsky, M.D., A. Kaplan, V.A. Terranova, and M.B. Hooten. (2026). A unified Bayesian framework for modeling measurement error in multinomial data. *Bayesian Analysis*, **21**: 453-483.

Ren, R., M.B. Hooten, T.L.J. Schafer, N.M. Calzada, B. Hoose, J.M. Womble, and S. Gende. (2026). A multistage Bayesian approach to fit spatial point process models. *Spatial Statistics*, **73**: 100975.

Robertson, E.C.N., T. Brown, S. Deitch, C.M. Bossu, E. Zavaleta, M.B. Hooten, and K. Ruegg. (2026). The genetic and morphological basis of local adaptation to elevational extremes in an alpine finch. *Ecology and Evolution*, **16**: e72962.

Gamba, D., M.L. Vahsen, T.M. Maxwell, N. Pirtel, S. Romero, J.J. Van Ee, A. Penn, A. Das, R. Ben-Zeev, O. Baughman, C.S. Blaney, R. Bodkins, S. Budha-Magar, S.M. Copeland, S.L. Davis-Foust, A. Diamond, R.C. Donnelly, P.W. Dunwiddie, D.J. Ensing, T.A. Everest, H. Hoitink, M.C. Holdrege, R.A. Hufbauer, S. Juzenas, J.M. Kalwij, E. Kashirina, S. Kim, M. Klisz, A. Klyueva, M. Langeveld, S. Lutfy, D. Martin, C.L. Merkord, J.W. Morgan, D.U. Nagy, J.P. Ott, R. Puchalka, L.A. Pyle, L. Rasran, B.G. Rector, C. Rosche, M. Sadykova, R.K. Shriver, A. Stanislavski, B.M. Starzomski, R.L. Stone, K.G. Turner, A.K. Urza, A. VanWallendael, C.-A. Wegenschimmel, J. Zweck, C.S. Brown, E.A. Leger, D.M. Blumenthal, M.J. Germino, L.M. Porensky, M.B. Hooten, P.B. Adler, J.R. Lasky. (2025). Local adaptation to climate has facilitated the global invasion of cheatgrass. *Nature Communications*, **16**: 10203.

Goel, N., C. Bossu, J.J. Van Ee, E. Zavaleta, K. Ruegg, and M.B. Hooten. (2025). Identifying genomic adaptation to local climate using mechanistic evolutionary models. *Methods in Ecology and Evolution*, **16**: 2448-2460.

Wright, W.J. and M.B. Hooten. (2025). Continuous-space occupancy models. *Biometrics*, **81**: ufaf055. [w/ discussion]. **Best paper in Biometrics Award 2025.**

Wright, W.J. and M.B. Hooten. (2025). Rejoinder to the discussion on “Continuous-space occupancy models.” *Biometrics*, **81**: ujaf058.

Lu, X., Y. Kanno, G. Valentine, M. Kulp, and M.B. Hooten. (2025). Regularized latent trajectory models for spatio-temporal population dynamics. *Journal of Agricultural, Biological and Environmental Statistics*, **30**: 683-699.

Wright, W.J., P.N. Neitlich, and M.B. Hooten. (2025). Clipped multiscale spatial processes for multivariate binary data. *Annals of Applied Statistics*, **19**: 2918-2936.

Schwob, M.R., M.B. Hooten, N. Calzada, and T.H. Keitt. (2025). Spatial hyperspheric models for compositional data analysis. *Annals of Applied Statistics*, **19**: 2644-2663.

Goljani, R., T. Brown, K. Leitner, G. Hong, K. Ruegg, M.B. Hooten, and E. Zavaleta. (2025). Hierarchical habitat selection for reconstructing past and present niches and distributions of data-limited species under climate change. *Diversity and Distributions*, **31**: e70128.

Lu, X., A. Kaplan, Y. Kanno, G. Valentine, J. Rash, and M.B. Hooten. (2025). Stochastic spatial stream networks for scalable inferences of riverscape processes. *Spatial Statistics*, **67**: 100902.

Vahsen, M.L., T.M. Maxwell, D.M. Blumenthal, D. Gamba, M.J. Germino, M.B. Hooten, J.R. Lasky, E.A. Leger, N. Pirtel, L.M. Porensky, S. Romero, J.J. Van Ee, S.M. Copeland, D.J. Ensing, P.B. Adler. (2025). Phenological sensitivity of *Bromus tectorum* genotypes depends on current and source environments. *Ecology*, **106**: e70025.

Goel, N., A.M. Liebhold, C. Bertelsmeier, M.B. Hooten, K.S. Korolev, and T.H. Keitt. (2025). A mechanistic statistical approach to infer invasion characteristics of human-dispersed species with complex life cycle. *Ecological Monographs*, **95**: e70003.

Van Ee, J.J., C.A. Hagen, D.C. Pavlacky Jr., D.A. Haukos, A.J. Lawrence, A.M. Tanner, B.A. Grisham, K.A. Fricke, L.G. Rossi, G.G. Beauprez, K.E. Kuklinski, R. Martin, M.D. Koslovsky, T.B. Rintz, and M.B. Hooten. (2024). Melded integrated population models. *Journal of Agricultural, Biological, and Environmental Statistics*, **30**: 769-799. **Top paper in JABES Award.**

Schwob, M.R., M.B. Hooten, and V. Narasimhan. (2024). Composite dyadic models for spatio-temporal data. *Biometrics*, **80**: ujae207.

Wikle, C.K., M.B. Hooten, W. Kleiber, and D.W. Nychka. (2024). Spatial statistics: Climate and the environment. *Spatial Statistics*, **63**: 100856.

Valentine, G.P., X. Lu, C.A. Dolloff, C.N. Roghair, J.M. Rash, M.B. Hooten, and Y. Kanno. (2024). Landscape influences on thermal sensitivity and predicted spatial variability among brook trout streams in the Southeastern USA. *River Research and Applications*, **40**: 1242-1255.

McDevitt-Gales, T., A.T. Degaetano, S. Elmendorf, J.R. Foster, H.S. Ginsberg, M.B. Hooten, S. LaDeau, K.M. McClure, S. Paull, E. Posthumus, I. Rochlin, and D. Grear. (2024). Partly Cloudy with a Chance of Mosquitoes: Developing a macroecological approach to forecasting mosquito populations and phenology under changing climates. *Ecosphere*, **15**: e70074.

Valle, D., N. Attias, J.A. Cullen, M.B. Hooten, A. Giroux, L. Gustavo, R. Oliveira-Santos, A.L.J. Desbiez, and R.J. Fletcher. (2024). Bridging the gap between movement data and connectivity analysis using the Time-Explicit Habitat Selection (TEHS) model. *Movement Ecology*, **12**: 19.

Leach, C.B., B. Weitzman, J. Bodkin, D. Esler, G.G. Esslinger, K.A. Kloecker, D. Monson, J.N. Womble, and M.B. Hooten. (2024). The dynamics of sea otter prey selection under population growth and expansion. *Ecosphere*, **15**: e70084.

Hui, F., Q. Vu, and M.B. Hooten. (2024). Spatial confounding in joint species distribution models. *Methods in Ecology and Evolution*, **15**: 1906-1921.

Lu, X., Y. Kanno, G. Valentine, G. Rash, and M.B. Hooten. (2024). Using multi-scale spatial models of dendritic ecosystems to infer abundance of a stream salmonid. *Journal of Applied Ecology*, **61**: 1703-1715.

Neitlich, P.N., W. Wright, E. Di Meglio, A.E. Shiel, C.J. Hampton-Miller, and M.B. Hooten. (2024). Mixed trends in heavy metal-enriched fugitive dust on National Park Service lands along the Red Dog Mine haul road, Alaska, 2006-2017. *PLoS One*, **19**: e0297777.

Hooten, M.B., M.R. Schwob, and D.S. Johnson. (2024). Geostatistical capture-recapture models. *Spatial Statistics*, **59**: 100817.

Eisaguirre, J.M., P.J. Williams, and M.B. Hooten. (2024). Rayleigh step-selection functions and connections to continuous-time mechanistic movement models. *Movement Ecology*, **12**: 14.

Valentine, G.P., X. Lu, E. Childress, C.A. Dolloff, N.P. Hitt, M.A. Kulp, B.H. Letcher, K.C. Pregler, J.M. Rash, M.B. Hooten, and Y. Kanno. (2024). Spatial asynchrony and cross-scale climate interactions in populations of a coldwater stream fish. *Global Change Biology*, **30**: e17029.

Lu, X., M.B. Hooten, A.M. Raiho, D.K. Swanson, C.A. Roland, and S.E. Stehn. (2023). Latent trajectory models for spatio-temporal dynamics in Alaskan ecosystems. *Biometrics*, **79**: 3664-3675.

Van Ee, J., C. Hagen, D. Pavlacky, K. Fricke, M. Koslovsky, and M.B. Hooten. (2023). Melding wildlife surveys to improve conservation inference. *Biometrics*, **79**: 3941-3953.

Schwob, M.R., M.B. Hooten, T. McDevitt-Gales. (2023). Dynamic population models with temporal preferential sampling to infer phenology. *Journal of Agricultural, Biological, and Environmental Statistics*, **28**: 774-791.

Lepak, J.M., B.M. Johnson, M.B. Hooten, B.A. Wolff, and A.G. Hansen. (2023). Predicting sport fish mercury contamination in heavily managed reservoirs: Implications for human and ecological health. *PLoS One*, **18**: e0285890.

Williams, P.J., X. Lu, H.R. Scharf, and M.B. Hooten. (2023). Embracing asymmetry in nature: How to account for skewness in ecological data. *Ecological Informatics*, **75**: 102185.

Hooten, M.B., M.R. Schwob, D.S. Johnson, and J.S. Ivan. (2023). Multistage hierarchical capture-recapture models. *Environmetrics*, **34**: e2799.

Eisaguirre, J.M., P.J. Williams, X. Lu, M.L. Kissling, P.A. Schuette, B.P. Weitzman, W.S. Beatty, G.G. Esslinger, J.N. Womble, and M.B. Hooten. (2023). Informing management of recovering predators and their prey with ecological diffusion models. *Frontiers in Ecology and the Environment*, **21**: 479-488

Leach, C.B., B.P. Weitzman, J. Bodkin, D. Esler, G.G. Esslinger, K.A. Kloecker, D. Monson, J.N. Womble, and M.B. Hooten. (2023). Revealing the extent of sea otter impacts on bivalve prey through multi-trophic monitoring and mechanistic models. *Journal of Animal Ecology*, **92**: 1230-1243.

Scharf, H.R., A. Raiho, S. Pugh, C.A. Roland, D.K. Swanson, S.E. Stehn, and M.B. Hooten. (2022). Multivariate Bayesian clustering using covariate-informed components with application to boreal vegetation sensitivity. *Biometrics*, **78**: 1427-1440.

Wright, W.J., P.N. Neitlich, A.E. Shiel, and M.B. Hooten. (2022). Mechanistic spatial models for heavy metal pollution. *Environmetrics*, **33**: e2760.

Wenger, S.J., E. Stowe, K. Gido, M. Freeman, Y. Kanno, N. Franssen, J.D. Olden, L. Poff, A. Walters, P. Bumpers, M. Mims, M.B. Hooten, and X. Lu. (2022). Simple statistical models can be sufficient for testing hypotheses with population time series data. *Ecology and Evolution*, **12**: e9339.

Okasaki, C., M.B. Hooten, and A.M. Berdahl. (2022). Source reconstruction for spatio-temporal physical statistical models. *Spatial Statistics*, **52**: 100707.

Lu, X., M.B. Hooten, A. Kaplan, J.N. Womble, and M.R. Bower. (2022). Improving wildlife population inference from aerial imagery data through entity resolution. *Journal of Agricultural, Biological, and Environmental Statistics*, **27**: 364-381.

Scharf, H.R., X. Lu, P.J. Williams, and M.B. Hooten. (2022). Constructing flexible, identifiable, and interpretable statistical models for binary data. *International Statistical Review*, **90**: 328-345.

Van Ee, J.J., J.S. Ivan, and M.B. Hooten. (2022). Community confounding in joint species distribution models. *Scientific Reports*, **12**: 12235.

Schafer, T.L.J., C.K. Wikle, and M.B. Hooten. (2022). Bayesian inverse reinforcement learning for collective animal movement. *Annals of Applied Statistics*, **16**: 999-1013.

Zimmerman, S., C. Aldridge, S. Oyler-McCance, and M.B. Hooten. (2022). Scale-dependent influence of the sagebrush community on genetic connectivity of the sagebrush obligate Gunnison sage-grouse. *Molecular Ecology*, **31**: 3267-3285.

Johnson, D.S., B.M. Brost, and M.B. Hooten. (2022). Greater than the sum of its parts:

Computationally flexible Bayesian hierarchical modeling. *Journal of Agricultural, Biological, and Environmental Statistics*, **27**: 382-400.

Kim, S., M.B. Hooten, T.L. Darden, and Y. Kanno. (2022). Linking male reproductive success to effort within and among nests in a co-breeding stream fish. *Ethology*, **128**: 489-498.

Raiho, A., H.R. Scharf, C.A. Roland, D.K. Swanson, S.E. Stehn, and M.B. Hooten. (In Press). Searching for refuge: A framework for identifying site factors conferring resistance to climate-driven vegetation change. *Diversity and Distributions*, **28**: 793-809.

Leach, C.B., P.J. Williams, J.M. Eisaguirre, J.N. Womble, M.R. Bower, and M.B. Hooten. (2022). Recursive Bayesian computation facilitates adaptive optimal design in ecological studies. *Ecology*, **103**: e03573.

Feuka, A.B., M.G. Nafus, A.A. Yackel Adams, L.L. Bailey, and M.B. Hooten. (2022). Endogenous and exogenous mechanisms affecting invasive reptile movement at multiple scales. *Movement Ecology*, **10**: 2.

Raiho, A., E.F. Nicklen, A. Foster, C.A. Roland, and M.B. Hooten. (2021). Bridging implementation gaps to connect large ecological datasets to complex models. *Ecology and Evolution*, **11**: 18271-18287.

Lepak, J.M., A.G. Hansen, M.B. Hooten, D. Brauch, and E.M. Vigil. (2021). Rapid proliferation of the parasitic copepod *Salmincola californiensis* on kokanee salmon in a large Colorado reservoir. *Journal of Fish Diseases*, **45**: 89-98.

Eisaguirre, J.M., P.J. Williams, X. Lu, M.L. Kissling, W.W. Beatty, G.G. Esslinger, J.N. Womble, and M.B. Hooten. (In Press). Diffusion modeling reveals effects of multiple release sites and human activity on a recolonizing apex predator. *Movement Ecology*, **9**: 34.

Banks, D.L. and M.B. Hooten. (2021). Statistical challenges in agent-based modeling. *The American Statistician*, **75**: 235-242.

Williamson, M.A., B.G. Dickson, M.B. Hooten, R.A. Graves, M.N. Lubell, and M.W. Schwartz. (2021). Accounting for incomplete reporting improves inference about private land conservation. *Conservation Biology*, **35**: 1174-1185.

Hooten, M.B., D.S. Johnson, and B.M. Brost. (2021). Making recursive Bayesian inference accessible. *The American Statistician*, **75**: 185-194.

McCaslin, H.M., A.B. Feuka, and M.B. Hooten. (2021). Hierarchical computing for hierarchical models in ecology. *Methods in Ecology and Evolution*, **12**: 245-254.

Lasky, J.R., M.B. Hooten, and P.B. Adler. (2020). What processes must we understand to forecast regional scale population dynamics? *Proceedings of the Royal Society, Series B*, **287**: 20202219.

Leach, C., J.A. Hoeting, K. Pepin, A. Eiras, M.B. Hooten, and C. Webb. (2020). Linking mosquito surveillance to dengue fever through Bayesian mechanistic modeling. *PLoS Neglected Tropical Diseases*, **14**: e0008868.

Hooten, M.B., C.K. Wikle, and M.R. Schwob. (2020). Statistical implementations of agent-based demographic models. *International Statistical Review*, **88**: 441-461.

Brost, B.M., M.B. Hooten, and R.J. Small. (2020). Model-based clustering reveals patterns in central place use of a marine top predator. *Ecosphere*, **11**: e03123.

Hooten, M.B., X. Lu, M.J. Garlick, and J.A. Powell. (2020). Animal movement models with mechanistic selection functions. *Spatial Statistics*, **37**: 100406.

Lu, X., P.J. Williams, M.B. Hooten, J.A. Powell, J.N. Womble, and M.R. Bower. (2020). Nonlinear reaction-diffusion process models improve inference for population dynamics. *Environmetrics*, **31**: e2604.

Hooten, M.B., S. Pugh, and C.A. Roland. (2020). Geary's contiguity ratio (Geary's c). *Wiley StatsRef: Statistics Reference Online*.

Christianson, K.R., B.M. Johnson, and M.B. Hooten. (2020). Compound effects of water clarity, inflow, wind, and climate warming on mountain lake thermal regimes. *Aquatic Sciences*, **82**: 6.

Tipton, J.R., M.B. Hooten, C. Nolan, R.K. Booth, and J. McLachlan. (2019). Predicting paleoclimate from compositional data using multivariate Gaussian process inverse prediction. *Annals of Applied Statistics*, **13**: 2363-2388.

Gerber, B.D., M.B. Hooten, C.P. Peck, M.B. Rice, J.H. Gammonley, A.D. Apa, and A.J. Davis. (2019). Extreme site fidelity as an optimal strategy in an unpredictable and homogeneous environment. *Functional Ecology*, **33**: 1695-1707.

Williams, P.J., W.L. Kendall, and M.B. Hooten. (2019). Selecting ecological models using multi-objective optimization. *Ecological Modelling*, **404**: 21-26.

Nolan, C., J. Tipton, R.K. Booth, M.B. Hooten, and S.T. Jackson. (2019). Comparing and improving methods for reconstructing peatland water table depth from testate amoebae. *The Holocene*, **29**: 1350-1361.

Hooten, M.B., J.M. Ver Hoef, and E.M. Hanks. (2019). Simultaneous autoregressive (SAR) model. *Wiley StatsRef: Statistics Reference Online*.

Scharf, H.R., M.B. Hooten, R.R. Wilson, G.M. Durner, T.C. Atwood. (2019). Accounting for phenology in the analysis of animal movement. *Biometrics*, **75**: 810-820.

Christianson, K.R., B.M. Johnson, M.B. Hooten, and J.J. Roberts. (2019). Estimating lake-climate responses from sparse data: an application to high elevation lakes. *Limnology and Oceanography*, **64**: 1371-1385.

Peterson, E.E., E.M. Hanks, M.B. Hooten, J.M. Ver Hoef, and M.-J. Fortin. (2019). Spatially structured statistical network models for landscape genetics. *Ecological Monographs*, **89**: e01355.

Williams, P.J., M.B. Hooten, G.G. Esslinger, J.N. Womble, J. Bodkin, and M.R. Bower. (2019). The rise of an apex predator following deglaciation. *Diversity and Distributions*, **25**: 895-908.

Ketz, A.C., T.L. Johnson, M.B. Hooten, and N.T. Hobbs. (2019). A hierarchical Bayesian approach for handling missing classification data. *Ecology and Evolution*, **9**: 3130-3140.

Hooten, M.B., H.J. Scharf, and J.M. Morales. (2019). Running on empty: Recharge dynamics from animal movement data. *Ecology Letters*, **22**: 377-389.

Dietze, M., A. Fox, L. Beck-Johnson, J.L. Betancourt, M.B. Hooten, C. Jarnevitch, T. Kiett, M. Kenney, C. Laney, L. Larsen, H. Loescher, C. Lunch, B. Pijanowski, J. Randerson, E. Reid, A. Tredennick, R. Vargas, K. Weathers, and E. White. (2018). Iterative near-term ecological forecasting: Needs, opportunities, and challenges. *Proceedings of the National Academy of Sciences*, **115**: 1424-1432

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#### **Awards/ Honors:**

- Best paper in Biometrics Award 2026  
Publication: Wright, W. and M.B. Hooten. (2025). Continuous-space occupancy models. *Biometrics*, 81: ujaf055.
- Knibbs Lecture 2025  
Statistical Society of Australia, Australian Capital Territory
- Best paper in JABES Runner Up Award 2025  
Publication: Van Ee, J.J., C.A. Hagen, D.C. Pavlacky Jr., D.A. Haukos, A.J. Lawrence, A.M. Tanner, B.A. Grisham, K.A. Fricke, L.G. Rossi, G.G. Beauprez, K.E. Kuklinski, R. Martin, M.D. Koslovsky, T.B. Rintz, and M.B. Hooten. (2024). Melded integrated population models. *Journal of Agricultural, Biological, and Environmental Statistics*, 30: 769-799.
- Distinguished Achievement Award 2022  
American Statistical Association, Section on Statistics and the Environment
- Superior Performance Award 2020  
U.S. Geological Survey
- Superior Performance Award 2019  
U.S. Geological Survey
- Wildlife Publication Award Shortlist for Authored Book 2019  
The Wildlife Society  
Publication: Hooten, M.B., D.S. Johnson, B.T. McClintock, and J.M. Morales. (2017). *Animal Movement: Statistical Models for Telemetry Data*. Chapman and Hall/CRC.
- Superior Performance Award 2018  
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- Superior Performance Award 2017  
U.S. Geological Survey
- ASA Fellow 2017  
American Statistical Association

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| ▪ President's Invited Lecture<br>The International Environmetrics Society Annual Meeting                                                                                                                                                               | 2016 |
| ▪ Superior Performance Award<br>U.S. Geological Survey                                                                                                                                                                                                 | 2016 |
| ▪ Outstanding Publication of the Year Award<br>Colorado State University, Warner College of Natural Resources<br>Publication: Hobbs, N.T. and M.B. Hooten (2015). Bayesian Models: A Statistical<br>Primer for Ecologists. Princeton University Press. | 2015 |
| ▪ Excellence in Science Award, Cooperative Research Units<br>U.S. Geological Survey                                                                                                                                                                    | 2015 |
| ▪ Superior Performance Award<br>U.S. Geological Survey                                                                                                                                                                                                 | 2015 |
| ▪ Superior Performance Award<br>U.S. Geological Survey                                                                                                                                                                                                 | 2014 |
| ▪ Young Investigator Award<br>American Statistical Association, ENVR Section                                                                                                                                                                           | 2014 |
| ▪ Superior Performance Award<br>U.S. Geological Survey                                                                                                                                                                                                 | 2013 |
| ▪ Superior Performance Award<br>U.S. Geological Survey                                                                                                                                                                                                 | 2012 |
| ▪ Superior Performance Award<br>U.S. Geological Survey                                                                                                                                                                                                 | 2011 |
| ▪ Researcher of the Year Award<br>USU-Department of Mathematics and Statistics                                                                                                                                                                         | 2010 |
| ▪ Researcher of the Year Award<br>USU-Department of Mathematics and Statistics                                                                                                                                                                         | 2009 |

## Editorial

### Experience:

- Associate Editor: Biometrics (2020-)
- Associate Editor: Environmetrics (2014-)
- Associate Editor: Journal of Agricultural, Biological, and Environmental Statistics (2012-13, 2017-)
- Associate Editor: Annals of Applied Statistics (2011-2021)
- Guest Editor: Special Issue in Spatial Statistics (2023-24)
- Guest Editor: Special Issue in Journal of Agricultural, Biological, and Environmental Statistics (2016-17)
- Guest Editor: Special Issue in Journal of Agricultural, Biological, and Environmental Statistics (2013-14)
- Subject Matter Editor: Ecological Applications (2018)

## Teaching

### Experience:

#### Workshops and Short Courses

|                                                                        |      |
|------------------------------------------------------------------------|------|
| ▪ Building Capacity in Bayesian Modeling for Ecologists (NSF), 10 days | 2024 |
| ▪ Building Capacity in Bayesian Modeling for Ecologists (NSF), 10 days | 2023 |
| ▪ Animal Movement Modeling Workshop (EURING), 1/2 day                  | 2023 |
| ▪ Building Capacity in Bayesian Modeling for Ecologists (NSF), 10 days | 2022 |
| ▪ Bayesian Statistical Inference and Practice (CPW), 2 days            | 2020 |
| ▪ R Spatial Data and Analysis (CSU), 1 day                             | 2020 |
| ▪ Animal Movement Modeling Workshop (US-IALE), 1 day                   | 2019 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Statistical Decision Theory (ASA Alaska Chapter Meeting)</li> <li>R Workshop (KSU), 1 day</li> <li>Animal Movement Modeling Workshop (ISEC), 1 day</li> <li>R Workshop (KSU), 1 day</li> <li>Spatio-Temporal Statistical Models in Practice (WNAR, anticipated), 1/2 day</li> <li>R Workshop for Wildlife Biologists (CSU-CCFWRU), 1 day</li> <li>Building Capacity in Bayesian Modeling for Ecologists (NSF), 10 days</li> <li>R Workshop for Wildlife Biologists (TWS-CMPS), 1 day</li> <li>Bayesian Decision Theory and Model Selection (ISEC), 1 day</li> <li>R Workshop (CSU-CCFWRU), 1 day</li> <li>Building Capacity in Bayesian Modeling for Ecologists (NSF), 10 days</li> <li>Parallel Computing for Ecologists and Evol. Biologists (CSU-CU), 1 day</li> <li>Building Capacity in Bayesian Modeling for Ecologists (NSF), 10 days</li> <li>R Workshop (CSU-CCFWRU), 1 day</li> <li>Building Capacity in Bayesian Modeling for Ecologists (NSF), 10 days</li> <li>Spatial Statistics using R Workshop (TWS), 1 day</li> <li>R Workshop for Fisheries Biologists (AFS-Western), 1 day</li> <li>Bayesian Models for Ecologists (USU - UCFWRU), 5 days</li> <li>R Short Course (CSU-CCFWRU), 1 day</li> <li>Bayesian Methods for Landscape Ecologists (US-IALE), 1 day</li> </ul> | 2019<br>2018<br>2018<br>2017<br>2017<br>2017<br>2016<br>2016<br>2016<br>2015<br>2015<br>2015<br>2014<br>2013<br>2013<br>2012<br>2012<br>2012<br>2011<br>2009 |
| The University of Texas at Austin, Dept. of SDS, Austin, TX. <ul style="list-style-type: none"> <li>Advanced Statistical Modeling and Applications I (SDS 383C/391P: Fall 2023, 2024, 2025, 2026)</li> <li>Bayesian Statistical Methods (SDS 384-7: Spring 2022, 2023, 2024, 2025, 2026)</li> <li>Elements of Statistics (SDS 320E: Fall 2022)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2021-                                                                                                                                                        |
| Colorado State University, Dept. of FWCB, Fort Collins, CO. <ul style="list-style-type: none"> <li>Hierarchical Models in Ecology (FW 680, Fall 2011; FW/STAT 673, Fall 2013, 2015, 2017, 2019)</li> <li>Fish, Wildlife, and Conservation Biology Graduate Faculty Seminar (FW 692, Spr. 2016)</li> <li>Readings on Bayesian Analysis of Ecological Models and Data (ECOL 592, Fall 2011)</li> <li>Independent Study, Wildlife Biology (FW 495, Fall 2014)</li> <li>Guest Lectures: STAT 501 (Fall 2011-2019), STAT 192 (Spring 2012-2015, Fall 2019), FW 696 (Fall 2018-2019)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2011-20                                                                                                                                                      |
| Utah State University, Dept. of Mathematics and Statistics, Logan, UT. <ul style="list-style-type: none"> <li>Applied Spatial Statistics (STAT 5410/6410, Fall 2006 - 2010)</li> <li>Statistics for Scientists (STAT 3000, Spring 2007, 2009, Fall 2007, 2008)</li> <li>Scientific Statistical Modeling: Directed Readings (STAT 6950, Spring 2007)</li> <li>Bayesian Statistics (STAT 6740, Spring 2008, 2010)</li> <li>Linear Regression and Time-Series (STAT 5100, Fall 2009 - 2010)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2006-10                                                                                                                                                      |
| University of Missouri, Statistics Dept., Columbia, MO. <ul style="list-style-type: none"> <li>Graduate Instructor <ul style="list-style-type: none"> <li>Statistical Methods for Agriculture Graduate Students (STAT 207)</li> <li>Probability and Statistics for Business Students (STAT 150)</li> </ul> </li> <li>Graduate Lecturer <ul style="list-style-type: none"> <li>Data Analysis for Graduate Students in Statistics (STAT 414)</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2002-04                                                                                                                                                      |
| University of Missouri, Forestry Dept., Columbia, MO. <ul style="list-style-type: none"> <li>Graduate Lecturer <ul style="list-style-type: none"> <li>Biometrics</li> <li>Geographic Information Systems</li> <li>Photogrammetry</li> <li>Remote Sensing</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1999-01                                                                                                                                                      |

### Post-Doctoral Fellows (Current):

- Myungsoo Yoo, Post-doctoral Fellow 2024-2026

### Post-Doctoral Fellows (Former):

- Justin Van Ee, Post-doctoral Fellow 2023-2026
- Alex Barth, Post-doctoral Fellow (co-advised with J. Casey) 2024-2025
- Nikunj Goel, Post-doctoral Fellow 2023-2025
- Clint Leach, Post-doctoral Fellow 2020-2024
- Xinyi Lu, Post-doctoral Fellow 2021-2023
- Ann Raiho, Post-doctoral Fellow 2019-2021
- Henry Scharf, Post-doctoral Fellow 2018-2019
- Perry Williams, Post-doctoral Fellow 2016-2018
- Brian Gerber, Post-doctoral Fellow 2016-2017
- John Tipton, Post-doctoral Fellow 2016-2017
- Kristin Broms, Post-doctoral Fellow 2013-2016
- Trevor Hefley, Post-doctoral Fellow 2015-2016
- Viviana Ruiz-Gutierrez, Post-doctoral Fellow 2013-2014
- Tabitha Graves, Smith Post-doctoral Fellow 2012-2014

### Graduate Students (Current):

- Brendan Allison (UT-Austin, PhD-Biology), Com. Member.
- Abraham Arbelaez (KSU, PhD-Statistics), Com. Member.
- Daniel Barreto (UT-Austin, PhD-Statistics), **Advisor**.
- Rachael Ren (UT-Austin, PhD-Statistics), **Advisor**.

### Graduate Students (Graduated):

- Michael Schwob (UT-Austin, PhD-Statistics), **Advisor**. 2025
- Wilson Wright (CSU, PhD-Statistics), **Co-Advisor** (w/ Dan Cooley). 2024
- Justin Van Ee (CSU, PhD-Statistics), **Co-Advisor** (w/ Matt Koslovsky). 2023
- George Valentine (CSU, MS-Ecology), **Co-Advisor** (w/ Yoichiro Kanno). 2023
- Xinyi Lu (CSU, PhD-Statistics), **Advisor**. 2021
- David Clancy (CSU, MS-Statistics), **Advisor**. 2019
- Henry Scharf (CSU, PhD-Statistics), **Advisor**. 2017
- Frances Buderman (CSU, PhD-Wildlife), **Advisor**. 2017
- Brian Brost (CSU, PhD-Ecology), **Advisor**. 2016
- John Tipton (CSU, PhD-Statistics), **Co-Advisor** (w/ Jean Opsomer). 2016
- Perry Williams (CSU, MS-Statistics), **Advisor**. 2015
- Shannon Kay (CSU, MS-Statistics), **Advisor**. 2015
- Alison Cartwright (CSU, MS-Statistics), **Co-Advisor** (w/ Jean Opsomer). 2013
- Ephraim M. Hanks (CSU, PhD-Statistics), **Advisor**. 2013
- Beth Ross (USU, PhD-Wildland Resources), **Co-Advisor** (w/ Dave Koons). 2013
- Martha Garlick (USU, PhD-App. Math), **Co-Advisor** (w/ Jim Powell). 2012
- Beth Ross (USU, MS-Statistics), **Advisor**. 2012
- Xiao Xiao (USU, MS-Statistics), **Advisor**. 2011
- Glenda Yenni (USU, MS-Statistics), **Advisor**. 2011
- Jess Anderson (USU, MS-Statistics), **Advisor**. 2011
- Mark Schmelter (USU, MS-Statistics), **Advisor**. 2011
- Ephraim M. Hanks (USU, MS-Statistics), **Advisor**. 2010
- Amanda R. Cangelosi (USU, MS-Statistics), **Advisor**. 2008
- Darl D. Flake (USU, MS-Statistics), **Advisor**. 2008

|                                                                                       |      |
|---------------------------------------------------------------------------------------|------|
| • Brandon Carter (UT-Austin, PhD-Statistics), Com. Member.                            | 2024 |
| • Hanna McCaslin (CSU, PhD-Ecology), Com. Member.                                     | 2023 |
| • Connie Okasaki (UW, PhD-Ecology), Com. Member.                                      | 2023 |
| • Andrew Manderson (Univ. of Cambridge, PhD-Statistics), Ext. Examiner                | 2022 |
| • Lachlan Griffin (QUT, MS-Statistics), Ext. Examiner                                 | 2021 |
| • Abigail Feuka (CSU, MS-Wildlife Biology), Com. Member.                              | 2021 |
| • Toryn Schafer (MU, PhD-Statistics), Com. Member.                                    | 2020 |
| • Francisco Peralta (Univ. of Cape Town, PhD-Statistical Ecology), External Examiner. | 2020 |
| • Ghulam Samad (CSU, PhD-Ecology), Com. Member.                                       | 2020 |
| • Clint Leach (CSU, PhD-Biology), Com. Member.                                        | 2019 |
| • Kyle Christianson (CSU, PhD-FWCB), Com. Member.                                     | 2019 |
| • Shawna Zimmerman (CSU, PhD-Ecology), Com. Member.                                   | 2018 |
| • Richard Glennie (Univ. of St. Andrews, Statistics), External Examiner.              | 2018 |
| • Clint Leach (CSU, MS-Statistics), Com. Member.                                      | 2017 |
| • Alison Ketz (CSU, PhD-Ecology), Com. Member.                                        | 2017 |
| • Yang Liu (UBC, PhD-Statistics), External Examiner.                                  | 2017 |
| • Zachary Weller (CSU, PhD-Statistics), Com. Member.                                  | 2017 |
| • Katy Warner (CSU, PhD-FWCB), Com. Member.                                           | 2016 |
| • Perry Williams (CSU, PhD-FWCB), Com. Member.                                        | 2015 |
| • Brian Gerber (CSU, PhD-FWCB), Com. Member.                                          | 2015 |
| • Kevin Blecha (CSU, MS-Ecology), Com. Member.                                        | 2015 |
| • Joe Northrup (CSU, PhD-Wildlife), Com. Member.                                      | 2015 |
| • Christian Roy (Univ. Laval, Canada, PhD-Ecology), External Examiner.                | 2015 |
| • Shane Siers (CSU, PhD-Ecology), Com. Member.                                        | 2014 |
| • Xiao Xiao (USU, PhD-Biology), Com. Member.                                          | 2014 |
| • Ann Raiho (CSU, MS-Ecology), Com. Member.                                           | 2014 |
| • Eric Gardunio (CSU, MS-FWCB), Com. Member.                                          | 2014 |
| • Glenda Yenni (USU, PhD-Biology), Com. Member.                                       | 2013 |
| • Aldo Compagnoni (USU, PhD-Wildland Resources), Com. Member.                         | 2013 |
| • Mark Schmelter (USU, PhD-Engineering), Com. Member                                  | 2013 |
| • Christy Meredith (USU, PhD-Wildland Resources), Com. Member.                        | 2012 |
| • Andrew Rayburn (USU, PhD-Wildland Resources), Com. Member.                          | 2011 |
| • John Lowry (USU, PhD-Wildland Resources), Com. Member.                              | 2010 |
| • Peter Sherick (USU, MS-Statistics), Com. Member.                                    | 2010 |
| • Audrey Smith (USU, MS-Mathematics), Com. Member.                                    | 2010 |
| • Tammy L. Wilson (USU, PhD-Wildland Resources), Com. Member.                         | 2010 |
| • Amanda Bakian (USU, MS-Statistics), Com. Member.                                    | 2008 |
| • Randy Larsen (USU, PhD-Wildland Resources), Com. Member.                            | 2008 |

#### **Undergraduates:**

- |                                                                                         |       |
|-----------------------------------------------------------------------------------------|-------|
| • Berkeley Ho (UT-Austin, Statistics and Data Sciences), Mentor and Research Supervisor | 2024- |
| • Katherine Millman (CSU, UG-Wildlife and Statistics), Honors Com. Member               | 2020  |

#### **Employees:**

- |                                          |           |
|------------------------------------------|-----------|
| • Nicolas Calzada, Data Scientist        | 2024-25   |
| • Michael Schwob, Statistical Technician | 2020      |
| • Christopher Peck, Research Associate   | 2017-2018 |
| • Jonathan Lewis, Research Associate     | 2015-2016 |
| • Joseph Halseth, Research Associate     | 2013-2015 |